TIRTHO ROY

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SUMMARY

Software and machine-learning engineer with a research-driven approach (B.S. Data Science and M.S. Mechanical Engineering, Iowa State University). Builds end-to-end systems—from TypeScript/React web applications and data-processing pipelines to large-scale machine learning, computer vision, and vision-language models—and is committed to turning research prototypes into reliable, well-tested, production-minded software. Co-first-authored work accepted at an IEEE/CVF Conference on Computer Vision and Pattern Recognition 2026 workshop.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL
Web & Full-Stack: React, web application development, REST APIs, responsive UI/UX
Machine Learning & AI: machine learning, deep learning (PyTorch), computer vision, vision-language models, large language models, multi-agent systems
Data & Infrastructure: data processing & pipelines, large-scale datasets, databases/SQL, cloud (AWS/GCP), Git, developer tooling

EDUCATION

Bachelor of Science in Data Science — Aug 2023 – Present
Iowa State University, Ames, IA

Master's of Science in Mechanical Engineering — Jan 2025 – Present
Iowa State University, Ames, IA

EXPERIENCE

Graduate Research Assistant — Jan 2025 – Present
Translational AI Center, Iowa State University — *Ames, IA*

- Lead M.S. thesis research on **agentic vision-language systems for cyber-agricultural applications** under Prof. Soumik Sarkar; co-first-authored work accepted at an **IEEE/CVF Conference on Computer Vision and Pattern Recognition 2026** workshop.
- Co-engineered **SAGE**, an autonomous crop-disease-diagnosis agent powered by the largest plant-disease image–symptom dataset to date (335 crops, 1,251 classes, ∼839K images); symptom grounding raised accuracy by **+16.2** points and generalizes to unseen crops with *no retraining*.
- Build large-scale LLM/VLM evaluation pipelines and reproducible benchmarks spanning agriculture, healthcare, and finance, turning research prototypes into deployable, rigorously tested systems.
- **Skills:** Python, PyTorch, multi-agent LLM/VLM orchestration, large-scale data engineering, retrieval & grounding, benchmark and metric design.

Undergraduate Research Assistant — Apr 2024 – Present
Translational AI Center, Iowa State University — *Ames, IA*

- Built a GPT-4o pipeline analyzing 430 Shark Tank pitches to quantify how market orientation and brand storytelling drive investor decisions; findings published in the *Journal of Business & Industrial Marketing*.
- **Skills:** Python, LLM prompting, human-AI annotation, statistical validation.

Undergraduate Software Developer — June 2024 – December 2024
Dept. of Agricultural & Biosystems Engineering, Iowa State University — *Ames, IA*

- Developed a chatbot web frontend for an agricultural research tool with Prof. Ryan McGhee, improving usability through iterative, user-tested design.
- **Skills:** web frontend development, responsive UI/UX design, user-centered design.

Undergraduate Research Volunteer Apr 2024 – May 2024
Dept. of Computer Science, Iowa State University *Ames, IA*

- Ran controlled probing experiments on how large language models resolve knowledge conflicts (conflicting context vs. parametric memory) under Prof. Qi Li.
- **Skills:** experimental design, LLM evaluation, data analysis.

PROJECTS

AgAdvisor — Agentic RAG for Pesticide-Label Guidance Live Demo
Translational AI Center / AIIRA, Iowa State University *(Iowa Soybean Association beta)*

- Built and deployed a multi-agent RAG assistant that answers pesticide-label questions grounded in **CDMS** labels, live as a public demo on Hugging Face Spaces; stack: FastAPI + LangGraph multi-agent orchestration, OpenAI embeddings + Qdrant vector search, GPT-4o, and a Streamlit/Gradio UI.
- **Industry connection:** developed in the Translational AI Center (AIIRA) and **beta-tested with the Iowa Soybean Association**, iterating directly on grower and agronomist feedback (wrong-herbicide bias, answer verbosity, mobile confidence badge, cold-start latency).
- Engineered **faithful, abstaining retrieval**—a product-disambiguating catalog plus abstention so the agent refuses rather than answer from the wrong herbicide label, with page-level citations: a concrete reliability guarantee for regulatory agrochemical guidance.
- Productionized with account authentication and per-user persistent history, per-user daily quotas to protect shared API keys, response caching, and a concurrency-safe shared vector store; a 62-test suite runs in CI.

PUBLICATIONS & PRESENTATIONS

SAGE: Scalable Agentic Grounded Evaluation for Crop Disease Diagnosis
IEEE/CVF Conference on Computer Vision and Pattern Recognition 2026 (CORE A*), Workshop on Emerging Directions in Data for Multimodal Foundation Models, Accepted *(Co-first author)*

- Co-built the largest plant-disease image–symptom dataset to date: 335 crops, 1,251 disease classes, and ~839K images with source-grounded, expert-validated symptom descriptions.
- Engineered an autonomous vision-language agent with stepwise diagnostic reasoning (anatomical context → symptom-guided narrowing → reference-image comparison → explainable trace); symptom grounding raised accuracy by +16.2 points on average and extends to new crops without retraining.
- Core component of M.S. thesis research.

SAGE: Scalable Agentic Grounded Evaluation for Crop Disease Diagnosis
AI Scientist Summer Workshop, Microsoft Research New England, Cambridge, MA, Poster *(Selected for presentation; co-first author)*

- Selected for poster presentation at the AI Scientist Summer Workshop hosted by Microsoft Research New England (August 4, 2026), presenting the SAGE crop-disease-diagnosis agent to the AI-for-science community.

bioMoR: Biology-Guided Mixture-of-Recursions for Effective Genomic Learning
Manuscript under review, 2026 *(Under review; second author)*

- Co-developed the first Mixture-of-Recursions framework for gene- and pathway-level omics learning, compressing high-dimensional expression inputs into marker-gene or Reactome pathway tokens and routing each token through an adaptive number of recursive steps in one weight-shared Transformer block.
- Injected biological structure at three sites — graph-based embedding smoothing, a structural attention bias over co-expression and pathway–pathway relations, and a zero-initialized graph-aware router that sets per-token recursion depth.

- Across eight single-cell and multi-omics benchmarks under a unified five-fold protocol, improved macro-F1 by 8.2 points and balanced accuracy by 7.1 points over the strongest biology-agnostic MoR baseline while using 75% fewer parameters and up to 58% fewer FLOPs than a non-recursive Transformer.

When Do Engineered Features Beat Raw Prices? A Walk-Forward Benchmark of Classical, Neural, and World-Model Predictors on a Global Equity Dataset
World Models @ Booth 2026 *(Under review)*

- Built a walk-forward backtesting framework benchmarking classical, neural, and world-model predictors on a global equity dataset.
- Isolated the conditions under which engineered features outperform raw price inputs across model families.

TRAPS: Therapeutic Response Analysis via Pathway-informed Stratification
arXiv preprint, 2026

- Built the first unified benchmark for pathway-guided cancer therapy-response modeling, evaluating three biologically informed architectures (BINN, GraphPath, PATH) under identical training and evaluation conditions.
- Engineered a multi-task pipeline over 2,622 TCGA patients across 5 cohorts (Reactome pathway-activity features), jointly predicting targeted therapy, radiation therapy, and 6-month survival; best model reached 0.92 AUROC on prostate targeted-therapy prediction at 11% class prevalence.

Improving the Safety and Trustworthiness of Medical AI via Multi-Agent Evaluation Loops
arXiv preprint, 2026

- Built a multi-agent iterative-refinement system pairing generator models (DeepSeek R1, Med-PaLM) with evaluator agents (LLaMA 3.1, Phi-4) to align medical LLM outputs to AMA ethics principles and a five-tier Safety Risk Assessment (SRA-5).
- Cut ethical violations by 89% and achieved a 92% risk-downgrade rate across 900 clinical queries spanning nine ethical domains.

An Agentic Framework for Mechanistic Therapeutic Reasoning in Oncology via Pathway-Grounded Tree-of-Thought Inference
North East AI Agents Day 2026, Accepted; ACM Conference on Bioinformatics, Computational Biology, and Health Informatics 2026 (CORE B), Posters

- Designed an agentic, pathway-grounded tree-of-thought inference pipeline that structures LLM reasoning over biological pathway knowledge for mechanistic therapeutic reasoning in oncology.

Intelligent Health Intervention System for Syndemic Management Among University Students: An AI-Driven Approach Using Reinforcement Learning
Association for the Advancement of Artificial Intelligence 2026 Spring Symposium Series, San Francisco, CA, Oral Presentation

- Co-developed a reinforcement-learning intervention engine for syndemic health management, modeling intervention timing and selection as a sequential decision problem.

Interpreting Transformers through the Lens of Physics and Dynamical Systems
American Physical Society Global Physics Summit 2026, Presentation *(First author)*

- First-authored an analysis interpreting transformer architectures through physics and dynamical-systems theory to explain their internal computation dynamics.

Leveraging Vision Language Models for Specialized Agricultural Tasks (AgEval Benchmark)
Proceedings of the 2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) (CORE A)

- Co-developed AgEval, a 12-task plant-stress phenotyping benchmark measuring zero- and few-shot in-context learning of state-of-the-art VLMs (Claude, GPT, Gemini, LLaVA).

- Demonstrated rapid domain adaptation: best-model F1 rose from 46.24% (zero-shot) to 73.37% (8-shot); a coefficient-of-variation metric exposed per-class gaps of 26–58%, and strategic example selection added +15.38% F1.

The Impact of Market Orientation and Brand Storytelling on Shark Tank Evaluations
Journal of Business & Industrial Marketing, 40(1), 265–280, 2025

- Built a GPT-4o analysis pipeline over 430 Shark Tank pitches to quantify how market orientation and brand storytelling drive investor decisions.
- Combined expert annotation with statistical validation to test and calibrate the LLM-derived predictions for a peer-reviewed B2B marketing study.

Enhancing Medical AI Safety Through Multi-Agent Evaluation and Iterative Refinement
The 16th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics 2025 (CORE B), Pennsylvania, USA, Poster

- Prototyped the multi-agent medical-LLM safety framework (generators DeepSeek R1 / Med-PaLM, evaluators LLaMA 3.1 / Phi-4) aligned to AMA principles and the SRA-5 protocol; later extended into the arXiv study above.

Federated In-Context Prompt Selection for Multi-Modal 3D Dental Imaging: A Theoretical Framework with Privacy-Preserving Guarantees
Oral and Dental Image Analysis Workshop 2025, held with the 29th International Conference on Medical Image Computing and Computer Assisted Intervention 2025 (CORE A), Poster

- Designed a privacy-preserving federated-learning framework for multi-modal 2D/3D dental imaging, featuring cross-modal prompt alignment, hierarchical multi-modal optimization, and Byzantine-resilient aggregation with differential privacy.

Emissai: LLM-Powered Analysis of Carbon Footprint
American Institute of Chemical Engineers Annual Meeting 2025, Presentation

- Built an LLM-powered tool to estimate and analyze carbon footprint from operational data.

Benchmarking Bioemu-1: Establishing a Baseline for Protein Structural Ensemble Prediction
American Institute of Chemical Engineers Annual Meeting 2025, Boston, USA, Poster

- Established baseline benchmarks for Bioemu-1 on protein structural-ensemble prediction, defining reproducible metrics for computational structural biology.

LlamaHealth: Assessing LLaMA's Performance in Multi-Class Healthcare Test Result Prediction
International Conference on Engineering Research, Innovation and Education 2025, Shahjalal University of Science and Technology, Sylhet, Bangladesh, Poster

- Benchmarked LLaMA on multi-class healthcare test-result prediction, assessing reliability for responsible medical-AI deployment.

LLM-based Insights into How Market Orientation and Brand Storytelling Affect Investors' Evaluation of Startup Pitches
Research Experiences for Undergraduates Summer Symposium 2025, Poster; Translational AI Center Day 2025, Poster & Flash Talk

- Presented the Shark Tank investor-evaluation study (GPT-4o over Shark Tank data) at the Research Experiences for Undergraduates Summer Symposium and as a Translational AI Center Day 2025 flash talk; mentored a peer researcher within the Translational AI Center.

INTERNSHIPS

Undergraduate Summer Research Intern – Translational AI Center May 2024 - July 2024

HONORS & AWARDS

Dean's High Impact Award for Undergraduate Research Fall 2025

Dean's List Fall 2024

Best-In-Show, George Washington Carver Poster Challenge 2025

Borlaug Poster Competition (1st Place, Undergraduate Division) 2025

Bangladesh–Sweden Trust Fund Travel Grant 2025

Iowa State University College of Liberal Arts and Sciences Travel Grant 2025

ACADEMIC SERVICE

Program Committee, Workshop on Secure and Trustworthy Quantum Machine Learning (SaTQuML) @ Conference on Neural Information Processing Systems 2026 (CORE A*)

Ethics Reviewer, Conference on Neural Information Processing Systems 2026 (CORE A*)

Reviewer, World Models @ Booth 2026

Reviewer, Context Beyond the Window Workshop @ Conference on Language Modeling (COLM) 2026 (top-tier)

Reviewer, Social Simulation Workshop (Social Sim '26) @ Conference on Language Modeling (COLM) 2026 (top-tier)

Ethics Reviewer, Conference on Neural Information Processing Systems 2025 (CORE A*) (Main Conference and Datasets & Benchmarks Tracks)

Reviewer, Rashomon Effect Workshop @ Association for the Advancement of Artificial Intelligence 2026 (CORE A*)

Reviewer, AI for Good Workshop @ International Conference on Machine Learning 2026 (CORE A*)

Reviewer, Computer Vision for Animals Workshop @ IEEE/CVF Conference on Computer Vision and Pattern Recognition 2026 (CORE A*)

Reviewer, AI for Critical National Infrastructure Workshop @ International Conference on Autonomous Agents and Multi-Agent Systems 2026 (CORE A*)

Reviewer, LatinX in AI Workshop @ Conference on Neural Information Processing Systems 2025 (CORE A*)

Reviewer, LatinX in Computer Vision Workshop @ IEEE/CVF Conference on Computer Vision and Pattern Recognition 2025 (CORE A*)

Reviewer, Second Workshop on Bangla Language Processing @ International Joint Conference on Natural Language Processing and Asia-Pacific Chapter of the Association for Computational Linguistics 2025

Reviewer, 4th International Conference on Computing Advancements (ICCA 2026)

Reviewer, 3rd International Conference on Machine Intelligence and Emerging Technologies (MIET 2026)